

DHANVANTARI: Autonomous Real-Time Multimodal AI Doctor & Clinical Triage System

A Statutory Telemedicine Practice Guidelines (2020) and DPDP Act (2023) Compliant Clinical Triage Platform with 3D Holographic WebGL Telehealth Interface and Neural VibeVoice Hindi Speech Synthesis

1. Executive Summary & Project Abstract

Dhanvantari is an end-to-end clinical AI physician triage and telehealth consultation system engineered to solve the dual challenges of urban-rural diagnostic disparities and preliminary patient triage throughput in India. Unlike conventional conversational chatbots, Dhanvantari combines a **LangGraph deterministic clinical state machine**, a **multimodal diagnostic extractor** for Chest X-Rays/CT scans and laboratory pathology reports, an automated **ICD-10 / NLEM 2022 grounded differential diagnosis engine**, a gated **DeepSeek-R1 671B deep reasoning layer**, and a real-time **3D Holographic WebGL neural core** that reacts organically to audio frequencies via Web Audio API. Crucially, the system is architected from the ground up to comply with India's **Telemedicine Practice Guidelines 2020** (mandating Registered Medical Practitioner human-in-the-loop sign-off) and the **Digital Personal Data Protection (DPDP) Act 2023** (cryptographic consent capture, purpose limitation, and local PII audit trails).

Key System Specifications & Benchmarks

Core Metric / Layer	Implementation Technology	Operational Functionality & Benchmark
State Machine	LangGraph + StateGraph	Deterministic clinical workflow preventing open-loop hallucination (Intake -> Safety -> RAG -> Diff-Dx -> Triage)
Fast Conversational LLM	Groq Cloud (LLaMA 3.3 70B / Qwen 3.6 27B)	Sub-second conversational triage & fluent Romanized Hinglish clinical formulation
Gated Deep Reasoner	DeepSeek-R1 (671B MoE via OpenRouter)	Multi-turn Chain-of-Thought (CoT) differential formulation for complex multimodal pathology
Doctor Voice Engine	tarun7r/vibevoice-hindi-1.5B + Neural TTS	Expressive human-like Indian physician persona with 100% full clinical depth speech alignment
3D Telehealth Stage	Three.js WebGL + Web Audio Analyser	Dynamic morphing neural core with 1,800+ orbiting photons reacting to voice pitch/loudness
Prescription Generator	ReportLab 5.0 + NLEM 2022 Formulary	Statutory electronic prescription generation with RMP digital signature & PDF download
Cloud Production	Render.com Docker + Cloudflare Edge	24/7 global availability with 100% mobile-responsive drawer & viewport optimization

2. End-to-End System Execution Flow & Methodology

The Dhanvantari platform executes through an asynchronous, event-driven multi-stage pipeline designed for safety, accuracy, and legal compliance at every step of patient interaction:

Stage	Phase & Component	Technical Execution Description & Safety Invariants
Phase 1	Patient Intake & DPDP Consent Capture	Captures patient identifiers and explicit, affirmative consent under India DPDP Act 2023. Generates cryptographically verifiable patient audit log in SQLite/PostgreSQL.
Phase 2	Multimodal Diagnostic Scan Ingestion	VisionExtractor ingests CXR, CT, MRI, and pathology report images. Employs SigLIP vision embeddings to extract quantitative metrics (e.g. Cardiothoracic ratio, opacity indices) and normal/abnormal status.
Phase 3	Deterministic Red-Flag Safety Gate	Hardcoded regex and semantic clinical rule-engine scans user input for 8 critical emergency categories (e.g. crushing chest pain, anaphylaxis, acute stroke). Immediately bypasses generative LLM and outputs emergency triage protocol.
Phase 4	RAG Knowledge Grounding & NLEM Lookup	Hybrid BM25 + Vector retriever pulls verified treatment guidelines from National List of Essential Medicines (NLEM 2022) and ICD-10 clinical codification dictionaries.
Phase 5	Gated Hybrid Intelligence (Groq + DeepSeek-R1)	Groq Cloud (LLaMA 3.3 70B) synthesizes real-time triage responses. If complex pathology, diagnostic ambiguity, or multimodal scans are present, the router gates execution to DeepSeek-R1 (671B) for Chain-of-Thought clinical reasoning.
Phase 6	VibeVoice 1.5B TTS & 3D WebGL Neural Core	Converts clinical advice into warm, fluent spoken audio via tarun7r/vibevoice-hindi-1.5B. Web Audio API AnalyserNode drives real-time vertex displacement and particle emission in the Three.js 3D Holographic Core.
Phase 7	RMP Doctor Review & Statutory PDF Prescription	Telemedicine Practice Guidelines 2020 loop: All escalated cases populate the RMP Doctor Dashboard. Upon physician verification and digital signature, ReportLab generates official downloadable PDF prescriptions.

Architectural Invariant: The system employs a Dual-Channel Clinical Dispatcher. The text channel delivers structured clinical markdown with differential diagnosis tables and precautions, while the neural voice channel translates the identical clinical guidance into natural, unhurried Hindi/Hinglish speech so patients receive 100% complete diagnostic context hands-free.

3. Technical Implementation & Subsystem Architecture

3.1 LangGraph State Machine & Node Architecture

The conversation flow is modeled as a directed acyclic graph (DAG) using LangGraph. The state object tracks patient demographics, turns history, multimodal image metadata, extracted symptoms, retrieved RAG context, DeepSeek-R1 reasoning traces, and statutory triage levels (Self-Care, See Doctor Soon, Urgent, Emergency).

3.2 Multimodal Diagnostic Extractor (VisionExtractor)

The VisionExtractor module supports radiographic (Chest X-Ray, CT) and pathological report ingestion. It performs image normalization, contrast enhancement, and zero-shot diagnostic inference using vision-language backends. The output is structured into clinical JSON containing: *modality*, *preliminary_impression*, *key_findings*, *extracted_metrics* (e.g. Hemoglobin, TLC, Platelet count with reference ranges), and *clinical_urgency* rating.

3.3 3D Holographic WebGL Visualizer & Web Audio Frequency Engine

Built using Three.js and WebGL, the 3D Telehealth Studio features a central morphing Icosahedron core surrounded by an outer wireframe cage and 1,800+ orbiting photon particles. Using the Web Audio API (AudioContext and AnalyserNode), real-time audio frequencies dynamically deform mesh vertex coordinates and modulate point lighting in real time:

Consultation State	Visualizer Color & Lighting	3D Mesh & Particle Dynamics
Patient Speaking	Vibrant Coral / Ruby Red (#EF4444)	Mesh vertices deform organically in response to voice frequency; particle orbit speeds accelerate.
Doctor Speaking (VibeVoice)	Luminous Celestial Teal / Cyan (#0D9488)	Emits harmonic sinusoidal wave ripples; golden photons orbit in synchronized radial patterns.
Analyzing Symptoms	Warm Amber / Golden Orange (#F59E0B)	Smooth sinusoidal breathing pulsation representing clinical differential evaluation.

4. Clinical Safety, Legal Compliance & Deployment

4.1 India Telemedicine Practice Guidelines (2020) Protocol

Under the regulations established by the National Medical Commission (NMC) and MoHFW, AI systems cannot independently prescribe medications. Dhanvantari enforces this through an architectural boundary:

- 1. Strict Prescription Draft Status:** All medication regimens generated by AI are flagged as 'DRAFT PENDING RMP APPROVAL'.
- 2. Schedule X Prohibition:** Controlled substances, habit-forming drugs, and Schedule X narcotics are strictly blocked by formulary filters.
- 3. RMP Review Dashboard:** Registered Medical Practitioners review patient consultation transcripts, modify dosages, and authenticate prescriptions via cryptographic signature before PDF generation.

4.2 Digital Personal Data Protection (DPDP) Act 2023 Architecture

Patient health records are categorized as sensitive personal information. Dhanvantari implements:

- **Granular Consent Modal:** Explicit opt-in consent recorded with immutable timestamps before triage session initiation.
- **Purpose Limitation:** Data is strictly bounded to real-time symptom assessment and RMP review.
- **Local Data Sovereignty:** SQLite/PostgreSQL storage with zero unauthorized third-party telemetry egress.

4.3 Production Deployment & Test Suite Verification

The system has been packaged with Docker and deployed to production cloud infrastructure on Render:

- **Live Cloud URL:** <https://dhanvantari-ai-doctor.onrender.com/>
- **RMP Doctor Review Portal:** <https://dhanvantari-ai-doctor.onrender.com/admin>
- **Interactive API Documentation:** <https://dhanvantari-ai-doctor.onrender.com/docs>
- **Automated Test Suite:** 6/6 test suites passed across red-flag detection, RAG retrieval, Hinglish synthesis, multimodal parsing, and prescription generation.

5. Project Conclusion & Internship Learning Outcomes

During this internship project, an enterprise-grade medical AI platform was conceptualized, built, and deployed. The project demonstrated mastery in modern AI engineering paradigms: **LangGraph deterministic state machines**, **multimodal medical vision inference**, **Three.js WebGL spatial audio interfaces**, **neural Hindi voice synthesis**, and **production Docker cloud deployment**. Dhanvantari serves as a blueprint for next-generation statutory-compliant telehealth infrastructure in India.

Student Intern: Saikat Biswas	Project Status: Completed & Live in Production
Role: Full-Stack AI Engineering Intern	Deployment: Render.com / Docker Cloud